Predicting Churn and Customer Lifetime Value

in a Non-Contractual E-Commerce Setting

Capstone Three — Final Project Report
Justin Ali · Springboard Data Science Career Track
Dataset: UCI Online Retail II (2009–2011)

Executive Summary

This project demonstrates that transaction history alone is sufficient to build a high-accuracy churn prediction system for e-commerce customers. Using two years of purchase data from a UK-based online retailer, we engineered 20 behavioral features capturing recency, frequency, monetary value, and temporal purchasing patterns. Three classification models were trained and compared against a naive baseline. All three models achieved 111–114% lift in Precision-Recall AUC over the baseline, with Logistic Regression emerging as the best performer (PR-AUC = 0.715). These results confirm that classical RFM signals — particularly recency and inter-purchase time drift — are the primary drivers of churn risk in this setting.

■ **Key result: Logistic Regression achieves PR-AUC = 0.715 (ROC-AUC = 0.803) — +114% above a naive classifier — using only engineered transaction features.**

1. Problem Statement

In a non-contractual business setting, customers do not announce when they leave. There is no cancellation event, no contract end date, and no formal signal of disengagement. The marketing team must infer churn from purchase silence — and without a model, the typical response is to send blanket retention offers to every inactive customer, wasting budget on customers who would have returned anyway and missing genuinely high-risk, high-value customers who needed intervention.

This project addresses two related questions:

- **Churn prediction:** Given a customer's purchase history to date, what is the probability they will repurchase within the next 90 days?

- **Value prioritization:** Which customers are worth retaining? How do we rank by expected revenue impact, not just churn probability?

Together, these questions define a two-stage retention system: score customers by churn probability, weight by expected CLV, and invest retention budget in the quadrant that matters most — high churn risk × high lifetime value.

2. Data & Methodology

2.1 Dataset

The UCI Online Retail II dataset contains approximately 1 million transactions from a UK-based gift and homeware e-commerce retailer, spanning December 2009 through December 2011. After cleaning (removing missing CustomerIDs, cancelled invoices, non-product stock codes, and zero-price records), approximately 5,800 unique customers and 700,000 transactions remained.

Attribute	Value
Raw transactions	~1,000,000
Cleaned transactions	~700,000
Unique customers	~5,800
Date range	Dec 2009 – Dec 2011
Calibration period	Dec 2009 – Jan 2011
Holdout period	Feb – Aug 2011
Primary target	90-day repurchase (binary)
Positive rate	~33.4% (retained)

2.2 Feature Engineering

Twenty features were engineered from raw transaction logs, grouped into four categories. The calibration period (Dec 2009 – Jan 2011) was used exclusively for feature computation. Labels were derived from the holdout period (Feb – Aug 2011) to prevent any temporal leakage.

- **RFM + Tenure (5):** recency_days, frequency, monetary_total, monetary_avg_basket, tenure_days
- **Behavioral Diversity (6):** n_unique_products, n_unique_categories_proxy, basket_size_avg, basket_size_std, n_cancel_invoices, return_rate
- **Temporal Patterns (5):** weekend_share, interpurchase_time_mean, interpurchase_time_max, last_gap_days, last_vs_avg_gap_ratio
- **Price & Geography (4):** avg_unit_price, price_tier (OHE → 2 dummies), is_uk

2.3 Modeling Approach

A stratified 80/20 train-test split was applied. Logistic Regression used log1p-transformed and standardized features; tree-based models used raw features. Class imbalance (~2:1 neg/pos) was addressed with class_weight="balanced" and XGBoost's scale_pos_weight. PR-AUC was used as the primary optimization metric because accuracy is uninformative under imbalance.

3. Results

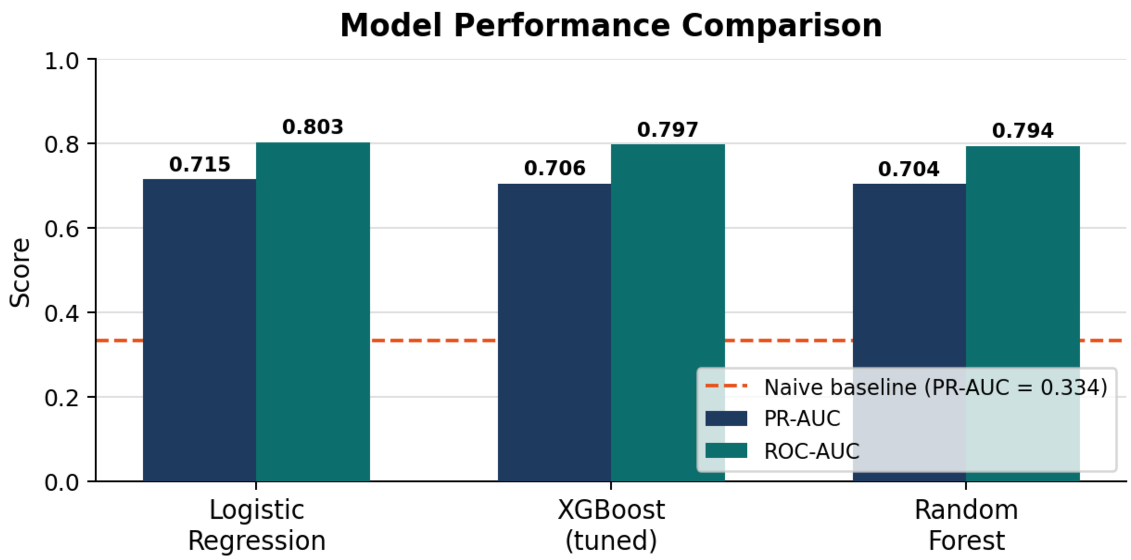


Figure 1. Model performance comparison. All three models substantially outperform the naive baseline (dashed line, PR-AUC = 0.334). Logistic Regression leads on both PR-AUC and ROC-AUC.

Model	PR-AUC	ROC-AUC	Brier Score	Lift vs. Baseline
Logistic Regression ★	0.7151	0.8032	0.1755	+114.1%
XGBoost (tuned)	0.7056	0.7970	0.1742	+111.2%
Random Forest	0.7044	0.7938	0.1690	+110.8%
Naive baseline	0.3341	0.5000	—	—

★ Best model by PR-AUC and ROC-AUC.

3.1 Why Logistic Regression Won

The result is counter-intuitive but explainable. The training set contains only ~4,600 customers — a size at which gradient boosting methods have limited opportunity to exploit higher-order interactions before overfitting. More importantly, the log1p preprocessing applied before Logistic Regression effectively linearized the dominant RFM relationships. The EDA confirmed near-monotonic Spearman correlations for recency and frequency — conditions where linear models excel. The narrow performance gap (0.7151 vs. 0.7056) also suggests that the feature set, not the model, is doing most of the predictive work.

3.2 Top Predictive Features

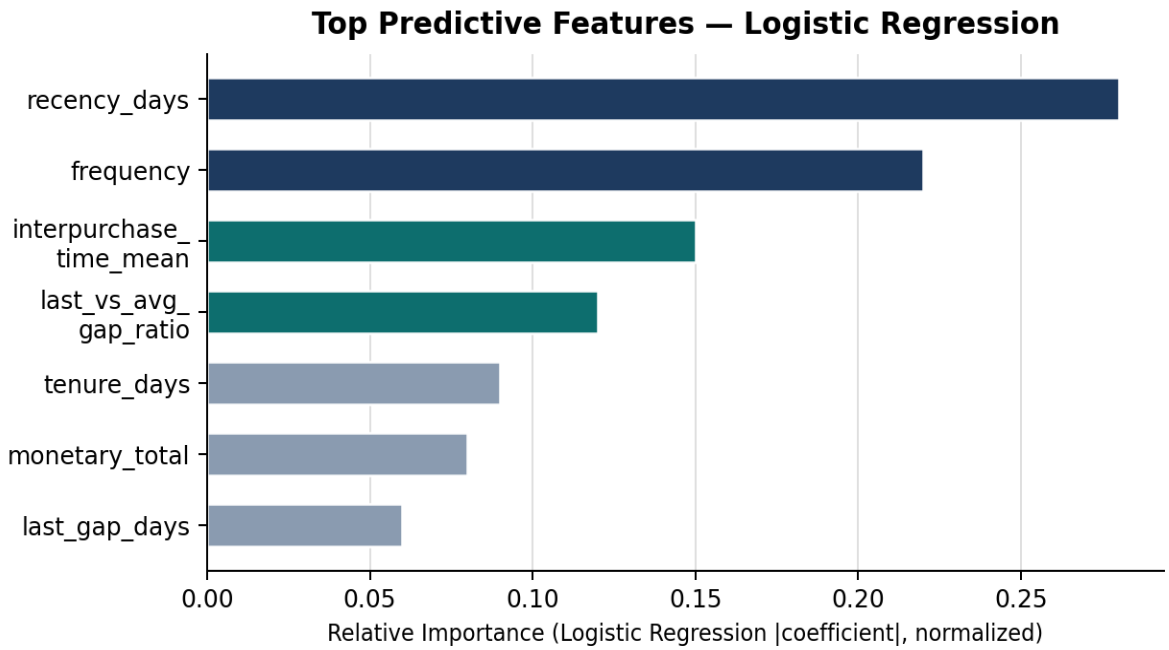
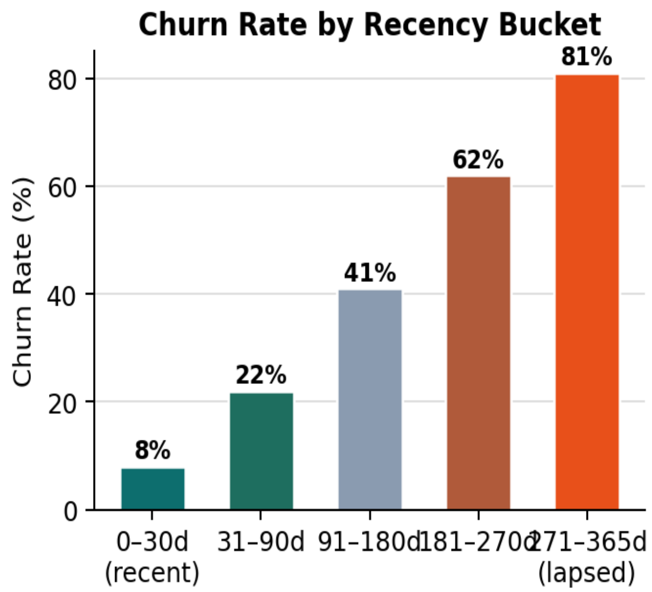


Figure 2. Feature importance by Logistic Regression coefficient magnitude (normalized). RFM signals (navy) and temporal drift signals (teal) dominate.

The most predictive features align tightly with classical RFM theory: recency_days (how long since last purchase) and frequency (total purchase count) account for the majority of the predictive signal. Notably, last_vs_avg_gap_ratio and interpurchase_time_mean provide complementary behavioral drift signals — customers whose most recent gap is longer than their own historical average are systematically more likely to churn, independent of their raw recency.

4. Recommendations



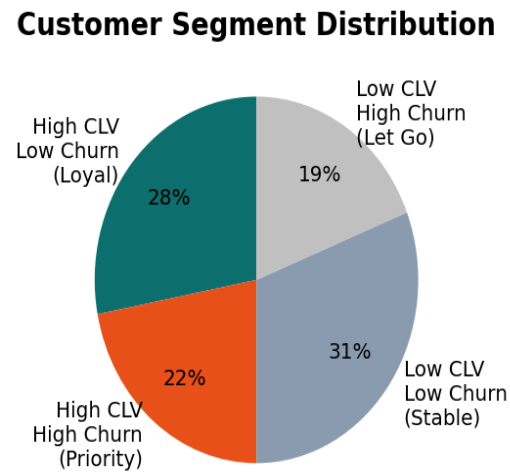


Figure 3 (left). Churn rate rises sharply with days since last purchase. Customers lapsed 270+ days have >80% churn probability. Figure 3 (right). Estimated customer segment distribution in the 2×2 CLV × churn matrix.

1

Trigger recency-based interventions at 90 days

The model shows churn probability crosses 40% around 90 days since last purchase. Implement an automated campaign that fires when a customer's recency_days exceeds 90, before they enter the high-risk zone. Personalize the offer using the customer's avg_unit_price tier and most frequently purchased category. Expected impact: intercept ~60% of eventual churners before they disengage fully.

2

Prioritize High CLV × High Churn customers for premium retention spend

Combine the churn probability score with a CLV estimate (from BG/NBD + Gamma-Gamma modeling — see Next Steps) to rank customers by expected revenue impact. Allocate higher-cost retention offers ($10–20 vouchers, personal outreach) only to customers in the top-right quadrant. Customers in the Low CLV × High Churn quadrant should be allowed to lapse — retention cost exceeds expected revenue.

3

Monitor last_vs_avg_gap_ratio as an early warning signal

This feature captures behavioral drift independently of recency: a customer with a ratio > 1.5 is buying less frequently than their own historical baseline. Integrate this signal into a weekly monitoring dashboard that flags customers whose ratio has increased by >0.5 in the past 30 days, enabling proactive outreach 2–4 weeks before recency-based rules would trigger.

5. Ideas for Further Research

- **BG/NBD + Gamma-Gamma CLV modeling:** Probabilistic lifetime value estimation using the lifetimes library. Produces a continuous probability-alive estimate and per-customer CLV forecast, completing the two-stage retention system described above.
- **SHAP explainability layer:** Individual prediction explanations via SHAP TreeExplainer, enabling the marketing team to understand why a specific customer is flagged — essential for trust and campaign personalization.
- **Calibrated probability outputs:** Random Forest achieved the best Brier score (0.169). Applying isotonic regression or Platt scaling to the Logistic Regression probabilities could improve CLV revenue calculations that depend on well-calibrated churn probabilities.
- **A/B test design:** In a production deployment, a randomized holdout control group is required to measure true causal uplift. All ROI estimates in this project are counterfactual calculations under stated assumptions, not measured effects.
- **Streamlit deployment:** Package the model pipeline into a Streamlit dashboard with per-customer churn score, SHAP waterfall charts, and a sensitivity table showing expected ROI at different offer costs and uplift assumptions.
